

server2 foundation operator manual

S04 provides a Rust host foundation for local validation. It serves operational health only. Weather/geocode, S3, provider adapters, capture jobs and production cutover are later tasks. Keep nginx and production routing on legacy.

Build and check

Prerequisites: pinned Rust 1.99.0 with rustfmt/clippy, Python 3.11+, and an isolated checkout. Cargo may download public dependencies; the binary never calls providers.

```
cd server2
cargo fmt --check
cargo clippy --locked --workspace --all-targets -- -D warnings
cargo test --locked --workspace
cargo build --locked --release
python3 tools/smoke.py --binary target/release/server2
```

The smoke starts two sequential release processes on ephemeral loopback ports. It checks GET/HEAD health, rejects public metrics, reads private counters, tests a 2-second incomplete-header timeout with elapsed bounds, socket-cap rejection/recovery, and both idle and partial-header SIGTERM shutdown. It also verifies that nonloopback metrics configuration fails startup. No legacy collectors run.

Configure and operate

Start server2/target/release/server2. Startup prints the two bound socket addresses. The default public listener is 127.0.0.1:3002; private metrics is 127.0.0.1:3003. Set SERVER2_BIND to change the public address. SERVER2_METRICS_BIND must stay loopback, including IPv6 ::1 if used. Invalid values and listener collisions fail startup.

Environment variable	Default	Allowed
SERVER2_WORKER_THREADS	2	1..64
SERVER2_BLOCKING_THREADS	4	1..64
SERVER2_CPU_LIMIT	2	1..blocking threads
SERVER2_INFLIGHT_LIMIT	40	1..1024
SERVER2_CACHE_BYTES	16777216	1..1073741824
SERVER2_CONNECTION_LIMIT	128	1..1024 public connections
SERVER2_CONNECTION_SECONDS	9	1..30 seconds maximum connection age

Metrics connections have a separate fixed limit 16. Socket limits bound slow headers and body draining; a connection expires at its maximum age even with keep-alive. This is an absolute connection lifetime, distinct from the later S10 per-request 9-second gateway deadline. These foundation limits require S10 assessment before cutover. The volatile cache is a single byte-bounded slot; keyed/sharded cache algorithms are S07. CPU permits remain reserved until blocking work ends after caller cancellation.

```
curl -i http://127.0.0.1:3002/health
curl -i http://127.0.0.1:3003/internal/metrics
```

Health returns 200 with exact body OK, text/html; charset=utf-8 and CORS *. HEAD returns 200, no body, Content-Length 2. Public /internal/metrics returns 404. Private metrics returns 200 and no-store, without CORS access, with only finite counters and declared limits. OPTIONS /health currently returns 405; legacy CORS preflight (204), session-cookie and complete middleware-header parity are deferred to S02 and not asserted here; S02 freezes the complete legacy oracle before route ports. SIGTERM/SIGINT drain both listeners for at most 5 seconds; a signal-triggered timeout logs a forced stop and exits successfully. Startup, signal-registration and unexpected serving failures remain unsuccessful. Tokio runtime shutdown then waits at most 1 second for blocking work, including error paths, giving a total shutdown bound of 6 seconds plus scheduling overhead. A CPU closure may outlive that wait until process exit; this does not claim a durable S3 or provider drain. Later tasks own their interruption guarantees.

Enforce placement before task completion

Declare owned directories and exact outside exceptions in config/tasks/<task>.json before editing and link the issue declaration. Use an exact reviewed task base:

```
cd server2
python3 tools/check_placement.py --root .. --declaration config/tasks/S04.json --base <exact-task-base>
python3 tools/test_placement.py
```

The gate scans full tracked server2 files, Cargo paths, symlinks and both rename ends. Rust checks cover include*/path/cfg_attr (all delimiters and comment separators) and direct read/read_to_string/read_dir/open/create/Path/PathBuf strings. Plain/raw and borrowed/parenthesized literals are supported; encoded or unsupported compiler arguments reject. Source-relative compiler paths are verified owned separately from runtime-CWD paths. Other Rust literals retain the legacy/sibling path guard, including Command arguments. Char/byte-char escape tokens and lifetimes stay aligned; ambiguous apostrophes reject incomplete.

Python executable-call AST, decoded JSON/TOML and shell/deploy paths are checked, including executable/shebang and .bash/.zsh scripts. Only validated task outside.path identities and exact test_placement.py self.write/fixture_symlink data roles are exempt; executable calls, runtime fields and invalid declarations still reject. Root Cargo and undeclared outside implementation reject. Git/JSON/TOML failures are incomplete checks. Binary executables, computed/aliased paths, custom loaders and macro-generated runtime access need human dependency/provenance review. These literal checks are a review aid, not an adversarial sandbox. Shared workflow permits checkout@v4/full history and one owned CI command, with a 30-minute timeout. It runs only on server2-related PRs. Remote branch protection is not configured by this task.

Actual usage capture

This capture comes from the 2026-10-06 release smoke on ephemeral ports. Its manifest records platform and binary/capture hashes. This is local foundation evidence; providers, deployment and full client parity were not exercised.

```
ACTUAL RELEASE-PROCESS SMOKE | 2026-10-06
python3.11 server2/tools/smoke.py --binary server2/target/release/server2
```

```
PASS invalid nonloopback metrics startup rejected
server2 public=127.0.0.1:53179 metrics=127.0.0.1:53180
GET public /health -> 200 OK; text/html; CORS *
HEAD public /health -> 200; empty body; Content-Length 2
GET public /internal/metrics -> 404
GET loopback metrics -> 200 no-store; counters: {"server2_cache_bytes": "0",
"server2_cache_limit_bytes": "16777216", "server2_cpu_inflight": "0", "server2_cpu_limit": "2",
"server2_health_requests_total": "2", "server2_inflight": "0", "server2_inflight_limit": "40",
"server2_rejected_total": "0"}
PASS public socket cap 2 rejects excess and recovers after release
PASS unfinished HTTP headers closed within 1.5..3.5s at 2-second lifetime
PASS SIGTERM drained both listeners; no provider calls
PASS SIGTERM with partial headers: logged clean forced stop within 4.5..7s
```